POSTDOCTORAL RESEARCH STATEMENT

Inference-Time Coordination of Foundation Models for Open-World 3D Scene Understanding

Applicant: Silas Kwabla Gah, Ph.D. Candidate | **Target Group:** Visual Computing & AI Lab (Prof. Dr. Matthias Nießner)

Position: Postdoctoral Researcher (TV-L E14) | Email: skgah001@st.ug.edu.gh | Web: silsgah.github.io | GitHub: github.com/silsgah

1. Research Vision & Alignment with TUM Visual Computing

Real-world embodied intelligence and spatial computing require 3D scene understanding systems that operate in an open-world: segmenting, localizing, and reasoning over novel categories without requiring catastrophic parameter retraining ($\Delta\theta = 0$). Prof. Dr. Matthias Nießner's Visual Computing and AI Group at the Technical University of Munich (TUM) has defined the global standard for this field through the pioneering creation of **ScanNet** and **ScanNet++**, alongside groundbreaking advances in neural implicit fields, 3D Gaussian Splatting (3DGS), and agentic 3D scene representations.

My doctoral dissertation, *"Visual Perception under Low Supervision: Inference-Time Coordination of Frozen Foundation Model Representations"* (University of Ghana, defending 2026/2027), was built directly upon the ScanNet and ScanNet++ benchmarks. My research establishes that heterogeneous foundation models (e.g., Segment Anything, CLIP, RegionPLC) already encode rich multi-modal visual evidence, but conventional fusion pipelines suffer from an information bottleneck caused by premature hard argmax collapse. At TUM, I aim to extend this foundation: unifying inference-time cross-view geometric arbitration with real-time neural radiance representations (3DGS) to realize scalable, training-free 3D/4D spatial intelligence.

2. Prior Research Breakthroughs on ScanNet & Foundation Models

Across my doctoral research, I have engineered and empirically validated novel inference-time arbitration frameworks across millions of spatial decision points:

A. Representation Before Weighting in Frozen 3D Fusion (*CVPR 2027*, Paper #9692, Under Review): A widespread implementation shortcut in multi-source 3D vision collapses continuous probabilities to hard Top-1 predictions before heterogeneous evidence can interact. In research under review at *CVPR 2027*, I isolated this representation bottleneck in frozen two-source 3D semantic fusion (RegionPLC + cross-view promptable masks) across 156 held-out ScanNet200 scenes. Formulating total-variation error bounds for normalized Top-k retention, I proved that semantic retention is *non-monotonic*: Top-20 retention achieves **35.49 Harmonic Mean (HM)**, significantly outperforming greedy Top-1 (28.47 HM) and Full distribution retention (34.87 HM). This proved that preserving plausible alternatives while controlling weak tails dominates downstream fusion weighting complexity.

B. Cross-Modal 2D-to-3D Correction via Geometric Consensus (*IVC*, Elsevier, Under Review): In 3D point cloud perception, naive uncalibrated lifting of promptable 2D masks (SAM/SAM3) creates severe overwrite degeneracy due to transient single-view occlusions and lens distortions. In work under review at *Image and Vision Computing*, I formulated a multi-view geometric consensus gate. While unconditional 2D overwrite degrades baseline performance from 31.56 to 28.04 HM, geometric ray consistency elevates performance to **33.68 HM on ScanNet200**, and on **ScanNet++ elevates novel-class mIoU from 16.16 to 32.20 (overall HM: 21.50 → 32.00)** without updating network weights.

C. Generalized Few-Shot Support Set Realism (*MVA*, Springer Nature, Under Review): Auditing generalized few-shot foundation model conditioning on frozen SAM3 and CLIP, I discovered that strong linguistic priors frequently overpower score-level visual support prototypes. Furthermore, I diagnosed an aspect-ratio letterboxing confound that silently shifts evaluation metrics by up to 39.96 HM points, establishing standardized coordinate protocols for foundation-model benchmarking.

D. High-Throughput Sparse Computing Systems: To conduct these extensive evaluations across ScanNet and ScanNet++ without access to multi-GPU enterprise server clusters, I engineered Coordinate (COO) sparse tensor serialization pipelines. This framework streams 21.4 GiB of high-dimensional scene representations at **2.08 million points/second** on standard memory budgets (RSS 719.5 MiB), enabling reproducible research at industry scale.

3. Proposed Postdoctoral Research Agenda at TU Munich

As a Postdoctoral Researcher in the Visual Computing & AI Group, I propose to drive two interrelated research frontiers aligned directly with ongoing projects in the Nießner Lab:

Direction 1: Streaming Open-Vocabulary 3D Gaussian Splatting (3DGS) with Geometric Arbitration. While recent works incorporate CLIP features into 3D Gaussian Splats, they typically rely on expensive per-scene offline optimization or suffer from spatial over-smoothing across semantic boundaries. I propose to develop an online, inference-time arbitration pipeline that lifts multi-view 2D zero-shot tokens directly into 3DGS primitives. By enforcing ray-consistent cross-view consensus and Shannon entropy gating, the system will dynamically filter spurious 2D hallucinations in real-time, achieving high-fidelity open-vocabulary 3D radiance field segmentation at 60+ FPS on interactive indoor scans.

Direction 2: Agentic 3D Scene Graphs & Continuous Spatiotemporal 4D Reasoning. Static 3D point cloud segmentations must be elevated into actionable representations for physical AI and robotics. Building upon the Nießner Lab's advances in hierarchical 3D scene understanding, I propose to formulate continuous spatiotemporal arbitration mechanisms that aggregate temporal multi-view streams into structured 3D scene graphs. By treating cross-view visual agreement as a self-supervised verification signal, embodied agents can autonomously resolve perceptual ambiguities and detect transient spatial state changes without human supervision.

4. Technical Capabilities & Contribution to the Lab Culture

I bring deep, hands-on engineering mastery across PyTorch, CUDA, C++, sparse matrix computation, and large-scale 3D data loaders, with extensive experience processing ScanNet, ScanNet++, and S3DIS. In addition to driving primary research papers targeting CVPR, ICCV, ECCV, and SIGGRAPH, I am committed to active lab citizenship: mentoring doctoral and master's students in experimental design, fostering open-source codebases, and contributing to TUM's world-leading visual computing ecosystem.